SUPPLEMENTARY MATERIAL
SINGLE CORRECT ANSWER

1. Let $f(x) = \dfrac{\tan^n x}{\sum_{r=0}^{2n} \tan^r x}$, $n \in N$, where $x \in \left[0, \dfrac{\pi}{2}\right)$

A) $f(x)$ is bounded and it takes both of it's bounds and the range of $f(x)$ contains exactly one integral point.
B) $f(x)$ is bounded and it takes both of it's bounds and the range of $f(x)$ contains more than one integral point.
C) $f(x)$ is bounded but minimum and maximum does not exists.
D) $f(x)$ is not bounded as the upper bound does not exist.

2. Two curves $C_1 : y = x^2 - 3$ and $C_2 : y = kx^2$, $k \in R$ intersect each other at two different points. The tangent drawn to C_2 at one of the points of intersection $A \equiv (a, y_1)$, $(a > 0)$ meets C_1 again at $B(1, y_2)$ $(y_1 \neq y_2)$. The value of 'a' is
A) 4 B) 3 C) 2 D) 1

3. A rectangle with one side lying along the x-axis is to be inscribed in the closed region of the xy plane bounded by the lines $y = 0$, $y = 3x$, and $y = 30 - 2x$. The largest area of such a rectangle is
A) $\dfrac{135}{8}$ B) 45 C) $\dfrac{135}{2}$ D) 90

4. Which of the following statement is true for the function

$$f(x) = \begin{cases} \sqrt{x} & x \geq 1 \\ x^3 & 0 \leq x \leq 1 \\ \dfrac{x^3}{3} - 4x & x < 0 \end{cases}$$

A) It is monotonic increasing $\forall\ x \in R$
B) $f'(x)$ fails to exist for 3 distinct real values of x
C) $f'(x)$ changes its sign twice as x varies from $(-\infty, \infty)$
D) function attains its extreme values at x_1 & x_2, such that $x_1, x_2 > 0$

5. Coffee is draining from a conical filter, height and diameter both 15 cms into a cylinderical coffee pot diameter 15 cm. The rate at which coffee drains from the filter into the pot is 100 cu cm /min. The rate in cms/min at which the level in the pot is rising at the instant when the coffee in the pot is 10 cm, is
A) $\dfrac{9}{16\pi}$ B) $\dfrac{25}{9\pi}$ C) $\dfrac{5}{3\pi}$ D) $\dfrac{16}{9\pi}$

6. Let f (x) and g (x) be two differentiable function in R and $f(2) = 8$, $g(2) = 0$, $f(4) = 10$ and $g(4) = 8$ then
A) $g'(x) > 4f'(x)\ \ \forall x \in (2, 4)$
B) $3g'(x) = 4f'(x)$ for at least one $x \in (2, 4)$
C) $g(x) > f(x)\ \ \forall x \in (2, 4)$
D) $g'(x) = 4f'(x)$ for at least one $x \in (2, 4)$

7. A horse runs along a circle with a speed of 20 km/hr . A lantern is at the centre of the circle . A fence is along the tangent to the circle at the point at which the horse starts . The speed with which the shadow of the horse move along the fence at the moment when it covers 1/8 of the circle in km/hr is
A) 20 B) 40 C) 30 D) 60

8. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false.
Statement-1: If $f : R \to R$ and $c \in R$ is such that f is increasing in $(c - \delta, c)$ and f is decreasing in $(c, c + \delta)$ then f has a local maximum at c. Where δ is a sufficiently small positive quantity.
Statement-2 : Let $f : (a, b) \to R$, $c \in (a, b)$. Then f can not have both a local maximum and a point of inflection at $x = c$.
Statement-3 : The function $f(x) = x^2 |x|$ is twice differentiable at $x = 0$.
Statement-4 : Let $f : [c - 1, c + 1] \to [a, b]$

be bijective map such that f is differentiable at c then f^{-1} is also differentiable at $f(c)$.

A) FFTF B) TTFT C) FTTF D) TTTF

9. Let $f: [-1, 2] \to R$ be differentiable such that $0 \le f'(t) \le 1$ for $t \in [-1, 0]$ and $-1 \le f'(t) \le 0$ $t \in [0, 2]$. Then

A) $-2 \le f(2) - f(-1) \le 1$

B) $1 \le f(2) - f(-1) \le 2$

C) $-3 \le f(2) - f(-1) \le 0$

D) $-2 \le f(2) - f(-1) \le 0$

10. If the function $f(x) = \dfrac{t + 3x - x^2}{x - 4}$, where 't' is a parameter has a minimum and a maximum then the range of values of 't' is

A) (0, 4) B) $(0, \infty)$ C) $(-\infty, 4)$ D) $(4, \infty)$

11. The function $S(x) = \int_0^x \sin\left(\dfrac{\pi t^2}{2}\right) dt$ has two critical points in the interval [1, 2.4]. One of the critical points is a local minimum and the other is a local maximum. The local minimum occurs at x =

A) 1 B) $\sqrt{2}$ C) 2 D) $\dfrac{\pi}{2}$

12. Read the following mathematical statements carefully:

I. A differentiable function ' f ' with maximum at $x = c \Rightarrow f''(c) < 0$.

II. Antiderivative of a periodic function is also a periodic function.

III. If f has a period T then for any $a \in R$.

$$\int_0^T f(x)\,dx = \int_0^T f(x+a)\,dx$$

IV. If $f(x)$ has a maxima at $x = c$, then 'f' is increasing in (c – h, c) and decreasing in (c, c + h) as $h \to 0$ for $h > 0$.

Now indicate the correct alternative.

A) exactly one statement is correct.

B) exactly two statements are correct.

C) exactly three statements are correct.

D) All the four statements are correct.

13. If the point of minima of the function, $f(x) = 1 + a^2x - x^3$ satisfy the inequality $\dfrac{x^2 + x + 2}{x^2 + 5x + 6} < 0$, then 'a' must lie in the interval

A) $\left(-3\sqrt{3}, 3\sqrt{3}\right)$

B) $\left(-2\sqrt{3}, -3\sqrt{3}\right)$

C) $\left(2\sqrt{3}, 3\sqrt{3}\right)$

D) $\left(-3\sqrt{3}, -2\sqrt{3}\right) \cup \left(2\sqrt{3}, 3\sqrt{3}\right)$

14. The radius of a right circular cylinder increases at a constant rate. Its altitude is a linear function of the radius and increases three times as fast as radius. When the radius is 1cm the altitude is 6 cm. When the radius is 6cm, the volume is increasing at the rate of 1Cu cm/sec. When the radius is 36cm, the volume is increasing at a rate of n cu. cm/sec. The value of 'n' is equal to

A) 12 B) 22 C) 30 D) 33

15. Consider $f(x) = |1 - x|$ $1 \le x \le 2$ and $g(x) = f(x) + b \sin \dfrac{\pi}{2} x$, $1 \le x \le 2$ then which of the following is correct?

A) Rolles theorem is applicable to both f, g and $b = \dfrac{3}{2}$

B) LMVT is not applicable to f and Rolles theorem if applicable to g with $b = \dfrac{1}{2}$

C) LMVT is applicable to f and Rolles theorem is applicable to g with $b = 1$

D) Rolles theorem is not applicable to both f, g for any real b.

16. Given that $f(x)$ is continuously differentiable on $a \le x \le b$ where $a < b$, $f(a) < 0$ and $f(b) > 0$, which of the following are always true?

(i) $f(x)$ is bounded on $a \le x \le b$.

(ii) The equation $f(x) = 0$ has at least one solution in $a < x < b$.

(iii) The maximum and minimum values of $f(x)$ on $a \le x \le b$ occur at points where $f'(c) = 0$.

(iv) There is at least one point c with a < c < b where f' (c) > 0.
(v) There is at least one point d with a < d < b where f' (c) < 0.
A) only (ii) and (iv) are true
B) all but (iii) are true
C) all but (v) are true
D) only (i), (ii) and (iv) are true

17. Suppose that f is a polynomial of degree 3 and that $f''(x) \neq 0$ at any of the stationary point. Then
A) f has exactly one stationary point.
B) f must have no stationary point.
C) f must have exactly 2 stationary points.
D) f has either 0 or 2 stationary points.

SUBJECTIVE TYPE

18. A window of fixed perimeter P (including the base of the arch) is in the form of a rectangle surrounded by a semicircle. The semicircular portion is fitted with coloured glass while the rectangular part is fitted with clear glass. transmits three times as much light per square meter as the coloured glass does. What is the ratio for the sides of the rectangle so that the window transmits the maximum light? (1991-4M)

19. Determine the points of maxima and minima of the function $f(x) = \frac{1}{8}\ln x - bx + x^2, x > 0$, where $b \geq 0$ is a constant.

20. Let $-1 \leq p \leq 1$. Show that the equation $4x^3 - 3x - p = 0$ has a unique root in the interval [1/2, 1] and identify it. (2001-5M)

21. Using Rolle's theorem, prove that there is at least one root in $(45^{1/100}, 46)$ of the polynomial
$P(x) = 51x^{101} - 2323(x)^{100} - 45x + 1035$. (2004-2M)

22. If $|f(x_1) - f(x_2)| \leq (x_1 - x_2)^2$, for all $x_1, x_2 \in R$. Find the equation of tangent to the curve y = f(x) at the point (1, 2). (2005-2M)

23. If p(x) be a polynomial of degree 3 satisfying p(-1) = 10, p(1) = -6 and p(x) has maxima at x= -1 and p′(x) has minima at x = 1. Find the distance between the local maxima and local minima of the curve. (2005-4M)

24. For a twice differentiable function f(x), g(x) is define d as g(x) = $f'(x)^2 + f''(x)$ f(x) on [a, e]. If for a < b < c < d < e, f(a) = 0, f(b) = 2, f(c) = -1, f(d) = 2, f(e) = 0 then find the minimum number of zeros of g(x). (2006-6M)

25. Let $a+b=4$ where $a<2$ and $g(x)$ be differentiable function. if $\frac{dg(x)}{dx} > 0$ for all x. The find whether $\int_0^a g(x)dx + \int_0^b g(x)dx$ increases or decreases as $(b-a)$ increases.

26. Let $A(p^2,-p), B(q^2,q), C(r^2,-r)$ be the vertices of the triangle ABC. A parallelogram AFDE is drawn with D, E and F on the line segments BC, CA and AB respectively. Using calculus show that the maximum area of such a parallelogram is $\frac{1}{4}(p+q)(q+r)(p-r)$.

27. A point P is given on the circumference of a circle of radius r. Chords QR are parallel to the tangent at P. Determine the maximum possible area of the traingle PQR.

28. Suppose f(x) is a function satisfying the following conditions
a. $f(0)=2,\ f(1)=1$
b. f has a minimum value at $x=\frac{5}{2}$
c. for all x,

$$f'(x)=\begin{vmatrix} 2ax & 2ax-1 & 2ax+b+1 \\ b & b+1 & -1 \\ 2(ax+b) & 2ax+2b+1 & 2ax+b \end{vmatrix}$$

where a, b are some constants. Determine the constants a, b and the function f(x).

29. **Circle $x^2+y^2=1$ cut X-axis at P &Q. Another circle with centre at Q & variable radius intersects the first circle at R above X- axis & line segement PQ at S. Find maximum area of QSR.**

30. **When travelling xkm/hr, a truck burns diesel at the rate of $\frac{1}{300}\left(\frac{900}{x}+x\right)$ liter per km. If the diesel oil costs 40 paise per litre and driver is paid Rs.1.5 perhour, find the steady speed that will minimise the total cost of the trip of 500 km.**

31. **A swimmer S is in the sea at a distance d km from the closest point A. The house of the swimmer is on the shore at a distance of L km from A. He can swim at a speed of u km/hr and walk at v km/hr. At what point on the shore he should land so he reaches houses in shortest time.**

32. **Tangent to curve $y=x-x^3$ at P meets curve again at Q. Locus at point of trisection. (other than point on y-axis)**

33. **Consider curve $5x^2-8xy+5y^2=4$ locate points on curve whose distance from origin is maxima & minima**

34. **Find polynomial function $f(x)$ of degree 6 which satisfy**

$$\lim_{x\to 0}\left\{1+\frac{f(x)}{x^3}\right\}^{1/x}=e^2$$

& has local maxima at $x=1$ & local min at $x=0$ and 2.

35. **Among all regular square pyramids of volume $36\sqrt{2}cm^3$. Find the dimensions of pyramid having least surface area.**

36. **If t be a real number satisfying $2t^3-9t^2-30-a=0$ value of parameter a for which $x+\frac{1}{x}=1$ gives six real and distinct values.**

37. **Function $\sqrt{ax^3+bx^2+cx+d}$ hs its non-zero local minimum & maximum values at $-2,2$. If 'a' is root of $x^2-x-6=0$ find a,b,c,d.**

38. **Two roads OA, OB intersect at 60^0 a car driver approaches O from A, $AO=800m$ at uniform speed at 20m/s simultaneously a runner starts running from O towards B at uniform speed 5m/s. Find time when car and runner are closest.**

39. **Jane is 2 miles of shore in a boat and wishes to reach a costal village 6mi down a straight line shoreline from the point nearest the boat. she can row 2mph and can walk 5mph, where should she land her boat to reach the village in least amount of time.**

40. **Minimum value of**

$$f(x)=\left||x+2|-2|x-2|\right|+|x|$$

<u>SUPLEMENTARY MATERAL</u>

<u>KEY</u>

<u>SINGLE CORRECT ANSWER</u>

1) A	2) B	3) C	4) C	5) D
6) D	7) B	8) A	9) A	10) C
11) C	12) A	13) B	14) D	15) C
16) D	17) D			

<u>SUBJECTIVE TYPE</u>

18. $6+\pi\ :6$

19. $\min \text{ at } x=\frac{1}{4}\left(b+\sqrt{b^2-1}\right),$

20. $\cos(\frac{1}{3}\cos^{-1}p)$

22. $y - 2 = 0$ 23. $4\sqrt{65}$ 24. 6

27) $\frac{1}{4}\left(3\sqrt{3}\right)r^2$ 28) $f(x)=\frac{1}{4}x^2-\frac{5}{4}x+2$

29) $\frac{4}{3\sqrt{3}}$ sq. units

HINTS

SUPPLEMENTARY MATERIAL

SINGLE CORRECT ANSWER

1. Let $\tan x = t$

$$\Rightarrow f(x)=\frac{t^n}{1+t+....+t^4+....+t^{2n}}=$$

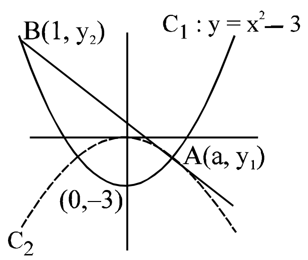
$$\frac{1}{\left(t^n+\frac{1}{t^n}\right)+\left(t^{n-1}+\frac{1}{t^{n-1}}\right)+.......+\left(t+\frac{1}{t}\right)+1}$$

$$\leq \frac{1}{2n+1}$$

[Equality holds at $x=\pi/4$]

also $f(0)=0 \Rightarrow$ range of $f(x)$ is $\left[0,\frac{1}{2n+1}\right]$

2.



Point $A(a, y_1)$ lies on C_1 and C_2

hence $y_1 = a^2 - 3$ and $y_2 = ka^2$

$\Rightarrow \quad a^2 - 3 = ka^2 \quad$ ....(1)

now $\quad y = kx^2 \Rightarrow \frac{dy}{dx}=2kx$

$$\therefore \quad \left.\frac{dy}{dx}\right]_{(a,y_1)}=2ka=\frac{y_2-y_1}{1-a}$$

(But $y_2 = 1 - 3 = -2$)

$$=\frac{-2-(a^2-3)}{1-a} \quad \Rightarrow 2ka=\frac{1-a^2}{1-a}=1+a$$

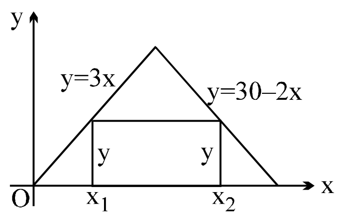
$2ka = 1 + a \quad$ .....(2)

Substituting $k=\frac{a^2-3}{a^2}$ from (1) in (2) we

get $\frac{2a(a^2-3)}{a^2}=1+a \quad \Rightarrow 2a^2-6=a+a^2$

$\Rightarrow a^2 - a - 6 = 0 \Rightarrow a = +3$, $a = -2$ (rejected)

3.



$A=(x_2-x_1)y$

$y = 3x_1$ and $y = 30 - 2x_2$

$$A(y)=\left(\frac{30-y}{2}-\frac{y}{3}\right)y$$

$6A(y) = (90 - 3y - 2y)y = 90y - 5y^2$

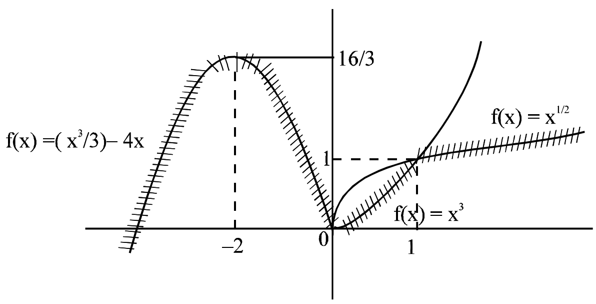
$6A'(y) = 90 - 10y = 0 \Rightarrow \quad y = 9$;

$A''(y) = -10 < 0$

$x_1 = 3$; $\quad x_2=\frac{21}{2}$

$$A_{max}=\left(\frac{21}{2}-3\right)9=\frac{15\cdot 9}{2}=\frac{135}{2} \quad \Rightarrow \quad (C)$$

4.



function is inc. in $(-\infty,-2)\cup(0,\infty)$

function is dec. in $(-2,0)$

$x=-2 \rightarrow$ local maxima

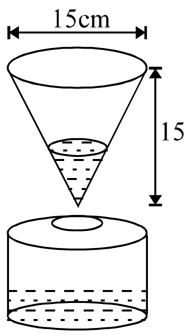
$x=0 \rightarrow$ local minima

Derivable $\forall x \in R - \{0,1\}$ –

$$\begin{cases} f'(0^+)=0,\ f'(0^-)=-4 \\ f'(1^+)=1/2,\ f'(1^-)=3 \end{cases}$$

Continuous $\forall x \in R$.

5.



For cylindrical pot $V = \pi r^2 h$

$$\frac{dV}{dt} = \pi\left[r^2\frac{dh}{dt} + h\cdot 2r\frac{dr}{dt}\right]$$

$(r = \text{constant}, \dfrac{dr}{dt} = 0)$

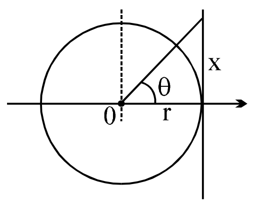
hence, $100 = \pi r^2 \dfrac{dh}{dt}$

$100 = \pi \cdot \dfrac{225}{4} \cdot \dfrac{dh}{dt} \quad (r = \dfrac{15}{2}\text{ cm})$

$\dfrac{dh}{dt} = \dfrac{400}{225\pi} = \dfrac{16}{9\pi}$ cm/min

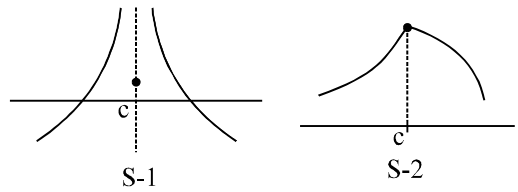
6. Consider $h(x) = g(x) - 4f(x)$, in $[2, 4]$
also $h(2) = g(2) - 4f(2) = -32; h(4) = -32$
$\Rightarrow h'(x) = 0$ for atleast one $x \in (2, 4)$ using Rolle's theorem

7.

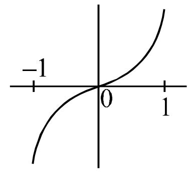


$\tan\theta = x/r \Rightarrow x = r\tan\theta$
$\Rightarrow dx/dt = r\sec^2\theta\,(d\theta/dt) = r\,\omega\sec^2\theta = v\sec^2\theta$
where $\theta = \pi/8$, $dx/dt = v\sec^2(\pi/4) =$
$2v = 40$ km/hr ; $\theta = 45°$

8.



S-1 : f must be continuous $\Rightarrow$ false
S-2 : $x = c$ is local max. & inflection point $\Rightarrow$ false



S-4

S-4 : Consider $f(x) = x^3$ at $x = 0$
$f^{-1}(x) = x^{1/3}$, not differentiable at $x = 0$

9. $0 \le f'(t) \le 1 \quad t \in [-1, 0]$

$\Rightarrow 0 \le \displaystyle\int_{-1}^{0} f'(t)\,dt \le 1$;

$0 \le f(0) - f(-1) \le 1 \quad$ ....(1)
now, $-1 \le f'(t) \le 0 \qquad t \in [0, 2]$

$$-2 \le \int_0^2 f'(t)\,dt \le 0$$

$-2 \le f(2) - f(0) \le 0 \quad$ ....(2)
(1) + (2)
$-2 \le f(2) - f(-1) \le 1$
Alternatively : use LMVT once in $[-1, 0]$ and then in $[0, 2]$

10. $f(x) = \dfrac{t + 3x - x^2}{x - 4}$;

$$f'(x) = \frac{(x-4)(3-2x) - (t + 3x - x^2)}{(x-4)^2}$$

for maximum or minimum, $f'(x) = 0$
$-2x^2 + 11x - 12 - t - 3x + x^2 = 0$
$-x^2 + 8x - (12 + t) = 0$
for one M and m,
$D > 0$
$64 - 4(12 + t) > 0$
$16 - 12 - t > 0 \Rightarrow 4 > t$ or $t < 4$

11. $S(x) = \int_0^x \sin\left(\frac{\pi t^2}{2}\right) dt$; $S'(x) = \sin\left(\frac{\pi x^2}{2}\right) = 0$

$\frac{\pi x^2}{2} = n\pi \Rightarrow x^2 = 2n$

$(1 \le x^2 \le 5.76$ as is given)

hence $n = 1$ or 2

$x = \sqrt{2}$ or $x = 2$; $S''(x) = \cos\left(\frac{\pi x^2}{2}\right). \pi x$

$S''(\sqrt{2}) < 0$ and $S''(2) > 0$

$\Rightarrow$ minima at $x = 2$

12. **I.** consider the function $f(x) = -x^4$, $f'(x) = -4x^3$ & $f''(x) = -12x^2$.

Here $f(x)$ has a maxima at $x = 0$ but $f''(0) = 0 \Rightarrow$ False.

II. $f(x) = \cos x + 1$ is periodic with period 2π

but $\int (\cos x + 1)\, dx = \sin x + x$ is not periodic.

III. $\int_0^T f(x+a)\, dx$, let $x + a = y$;

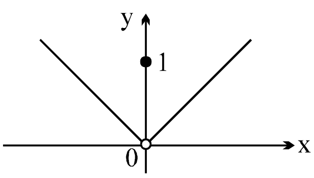
$$\int_a^{a+T} f(y)\, dy = \int_a^0 f(y)\, dy + \int_0^T f(y)\, dy + \int_T^{a+T} f(y)\, dy$$

consider $\int_T^{a+T} f(y)\, dy$; $y = T + v$

$$\int_0^a f(v+T)\, dv = \int_0^a f(v)\, dv.$$

Hence $\int_0^T f(x)\, dx = \int_0^T f(x+a)\, dx \Rightarrow$ True.

IV.



$$f(x) = \begin{cases} x & \text{if } x > 0 \\ 1 & \text{if } x = 0 \\ -x & \text{if } x < 0 \end{cases}$$

This has a maxima at $x = 0$ however this does not satisfy conditions stated in the problem $\Rightarrow$ False

13. $f'(x) = 0 \Rightarrow x = \frac{a}{\sqrt{3}}$ or $-\frac{a}{\sqrt{3}}$

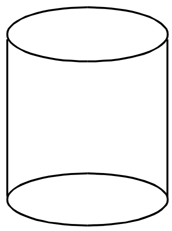
$f''(x) = -6x$

Case I if $a > 0 \Rightarrow x = -\frac{a}{\sqrt{3}}$ is minima

Case II if $a < 0 \Rightarrow x = \frac{a}{\sqrt{3}}$ is minima

put $x = \frac{a}{\sqrt{3}}$ and then $x = -\frac{a}{\sqrt{3}}$ in the given inequality to get the result

14.



$\frac{dr}{dt} = c$ and $h = ar + b$;

also $\frac{dh}{dt} = 3\frac{dr}{dt}$ (given)

$\therefore \quad a\frac{dr}{dt} = 3\frac{dr}{dt} \Rightarrow a = 3$

hence $h = 3r + b$

when $r = 1$; $h = 6 \Rightarrow 6 = 3 + b$

$\Rightarrow b = 3$

$\therefore \quad h = 3(r+1)$

$V = \pi r^2 h = 3\pi r^2(r+1) = 3\pi(r^3 + r^2)$

$\frac{dV}{dt} = 3\pi(3r^2 + 2r)\frac{dr}{dt}$

where $r = 6$; $\frac{dV}{dt} = 1$ cc/sec

$\therefore \; 1 = 3\pi(108 + 12)\frac{dr}{dt} \Rightarrow 360\pi\frac{dr}{dt} = 1$

again when $r = 36$, $\frac{dV}{dt} = n$

$n = 3\pi\,((3.36)^2 + 2.36)\,\frac{dr}{dt}$

$n = 3\pi\,.\,36\,(110)\,.\,\frac{1}{360\pi}$

$n = 33 \quad \Rightarrow \quad$ (D)

15. $f(x) = x - 1,\ 1 \le x \le 2$

$g(x) = x - 1 + b\sin\frac{\pi}{2}x,\ \ 1 \le x \le 2$

$f(1) = 0;\ f(2) = 1 \ \Rightarrow$ Rolle's theorem is not applicable to 'f' but LMVT is applicable to f.
($\because\ x - 1$ is continuous and differentiable in [1, 2] and (1, 2) respectively)
Now $g(1) = b;\ g(2) = 1$ and

Function $x - 1, \sin\frac{\pi}{2}x$ are both continuous in [1, 2] and (1, 2)
$\therefore$ For Rolle's theorem to be applicable to g.
We must have b = 1

16. (i) This statement is true, every continuous function is bounded on a closed interval
(ii) True again, by Intermediate Value Theorem
(iii) Not true, because maximum and / or minimum values could also occur at a or b, without the derivatives being 0.
(iv) True. By the Mean Value Theorem there exist a point between a and b where the derivative is exactly $\frac{f(b) - f(a)}{b - a}$, a clearly positive value.
(v) Not always true, for example the function might be strictly increasing guarenteeing the derivative to be always positive.
Thus the true statements are (i), (ii) and (iv) and the correct answer is (D)

17. The derivative of a degree 3 polynomial is a quadratic. This must have either 0, 1 or 2 roots. If this has precisely one root, then this must be repeated.
Hence we have $f'(x) = m(x - \alpha)^2$, where α is repeated root and $m \in R$. So our original function f has a critical point at $x = \alpha$.
Also $f''(x) = 2m(x-\alpha)$, in which case $f''(\alpha) = 0$. But we are told that the 2nd derivative is non zero at critical point. Hence there must be either 0 or 2 critical point.

18. light transmission rate (per square metre) of the coloured glass be L and Q be the total amount of transmitted light.
Then, $Q = 2ab\,(3L) + 1/2\ \pi b^2\,(L)$
$Q = L/2\ \{\pi b^2 + 12ab\}$
$Q = L/2\ \{\pi b^2 + 6b\,(K - 4b - \pi b)\}$
$Q = L/2\ \{6Kb - 24b^2 - 5\pi b^2\}$

$\frac{dQ}{db} = \frac{L}{2}\ \{6K - 48b - 10\pi b\} = 0$

$\Rightarrow b = \frac{6K}{48 + 10\pi}$ ... (ii)

and $\frac{d^2Q}{db^2} = \frac{L}{2}\ \{-48 + 10\pi\}$ La

Thus, Q is maximum and from (i) and (ii),
$(48 + 10\pi)\,b = 6K$ and $K = 2a + 4b + \pi b$
$\Rightarrow (48 + 10\pi)\,b = 6\,\{2a + 4b + \pi b\}$

Thus, the ratio $= \frac{2b}{a} = \frac{6}{6 + \pi}$

19. f(x) is a differentiable function for x > 0.
Therefore, for maximum or minima $f'(x) = 0$ must satisfy.

We have $f(x) = \frac{1}{8}\ln x - bx + x^2, x > 0$

$\Rightarrow\ f'(x) = \frac{1}{8}.\frac{1}{x} - b + 2x$ for $f'(x) = 0$

$(4x - b)^2 = (b - 1)(b + 1)$, $\{b \ge 0$ (given)$\}$
Case I: $0 \le b < 1$, (1) has no solution.
Since R.H.S. is negative in this domain and L.H.S. is positive

Case II: $b = 1, \Rightarrow x = \frac{1}{4}$ is the only solution.

when b = 1

$$f'(x) = \frac{1}{8x} - 1 + 2x = \frac{2}{x}\left(x^2 - \frac{1}{2}x + \frac{1}{16}\right) = \frac{2}{x}\left(x - \frac{1}{4}\right)^2$$

$f'(x)$ has no change of sign in left and right of $x = 1/4$.
Case III. b > 1, then

$$f'(x) = \frac{1}{8x} - b + 2x = \frac{2}{x}(x - \alpha)(x - \beta)$$

where $\alpha < \beta$ and $\alpha = \frac{1}{4}\left(b - \sqrt{b^2 - 1}\right)$ and

$\beta = \frac{1}{4}\left(b + \sqrt{b^2 - 1}\right)$. From sign scheme it is

clear that $f'(x)\begin{cases} > 0 \text{ for } 0 < x < \alpha \\ < 0 \text{ for } \alpha < x < \beta \\ > 0 \text{ for } x > \beta \end{cases}$

By the first derivative test, f(x) has a maximum at $x = \alpha$ and f(x) has a minimum at $x = \beta$

20. Given that $-1 \leq p \leq 1$

Consider $f(x) = 4x^3 - 3x - p$

Now, $f(1/2) = \frac{1}{2} - \frac{3}{2} - p = -1 - p \leq 0$,

(as $p \geq -1$)

also $f(1) = 4 - 3 - p = 1 - p \geq 0$, (as $p \leq 1$)

$\therefore$ f(x) has atleast one real root between [1/2, 1]

Also, $f'(x) = 12x^2 - 3 > 0$, on [1/2, 1]

$\Rightarrow f'(x)$ increasing on $\left[\frac{1}{2}, 1\right]$

$\Rightarrow$ f has only one real root between [1/2, 1]

To find a root, we observe f(x) contains $4x^3 - 3x$ which is multiple angle formula for $\cos 3\theta$ $\quad\therefore$ we put $x = \cos\theta$

$\Rightarrow 4\cos^3\theta - 3\cos\theta - p = 0$

$\Rightarrow p = \cos 3\theta$ or $\theta = \frac{1}{3}\cos^{-1}(p)$

$\therefore$ Root is $\cos\left[\frac{1}{3}\cos^{-1}(p)\right]$

21. Consider the function $g(x) =$

$$\frac{x^{102}}{2} - \frac{2323x^{101}}{101} - \frac{45}{2}x^2 + 1035x$$

Now g(x) is a polynomial function

it is continuous and differentiable in the given interval, also $g(46) = g(45^{1/100})$

Rolle's theorem is applicable here.

By Rolle's theorem $g'(c) = 0$ for some $c \in (45^{1/100}, 46)$ or

$p(x) = 0$ for some $c \in (45^{1/100}, 46)$

Hence the given expression has a solution in the given interval.

22. As $|f(x_1) - f(x_2)| \leq (x_1 - x_2)^2 \;\; \forall x_1, x_2 \in R$

$\Rightarrow |f(x_1) - f(x_2)| \leq |x_1 - x_2|^2$ $\{$as $x^2 = |x|^2\}$

$\therefore \left|\frac{f(x_1) - f(x_2)}{x_1 - x_2}\right| \leq |x_1 - x_2|$

Taking limit $x_1 \to x_2$, we get

$\lim\limits_{x_1 \to x_2}\left|\frac{f(x_1) - f(x_2)}{x_2 - x_2}\right| \leq \lim\limits_{x_1 \to x_2} |x_1 - x_2|$

$\Rightarrow |f'(x_1)| \leq 0 \;\forall x_1 \in R$

$\therefore |f'(x)| \leq 0$, which shows $|f'(x)| = 0$

$\{$as modulus is non negative or $|f'(x)| \geq 0\}$

$\therefore f'(x) = 0$ or f(x) is constant function

$\Rightarrow$ Equation of tangent at (1, 2) is

$\frac{y-2}{x-1} = f'(x)$ or $y - 2 = 0$ (as $f'(x) = 0$)

$\Rightarrow y - 2 = 0$ is required equation of tangent.

23. $p''(x) = k(x-1) \Rightarrow p'(x) = k\left(\frac{x^2}{2} - x\right) + k_1$

$\because p'(-1) = 0 \Rightarrow k_1 = \frac{-3k}{2}$

$\Rightarrow p'(x) = k\left(\frac{x^2}{2} - x - \frac{3}{2}\right)$

Now $p(x) = k\left(\frac{x^3}{6} - \frac{x^2}{2} - \frac{3x}{2}\right) + k$

Now by using $p(-1) = 10$, $p(1) = -6$

We get $k = 6$, $k_1 = 5$

Now $\quad P(x) = x^3 - 3x^2 - 9x + 5$

$p'(x) = 3(x^2 - 2x - 3) = 0 \Rightarrow x = -1, 3$ and

$p^2(3) > 0$ and $p''(-1) < 0$

$\Rightarrow (-1, 10)\ (3, -22)$ are the points of y maxima and minima & distance between them

$= \sqrt{1040} = 4\sqrt{65}$

24. Let, $g(x) = \frac{d}{dx}[f(x) \cdot f'(x)]$

to get the zero of g(x) we take function

$h(x) = f(x).f'(x)$

between any two roots of h(x) there lies at least one root of $h'(x) = 0$

$\Rightarrow g(x) = 0 \Rightarrow h(x) = 0 \Rightarrow f(x) = 0$ or $f'(x) = 0$

If $f(x) = 0$ has 4 minimum solutions.

$f'(x) = 0$ has 3 minimum solutions.

$h(x) = 0$ has 7 minimum solutions.

$\Rightarrow h'(x) = g(x) = 0$ has 6 minimum solutions.

25. given $a + b = 4$

$b - a = 4 - a - a = 4 - 2a$

$$\int_0^a g(x)dx + \int_0^b g(x)dx =$$

$$\int_0^a g(x)dx + \int_0^{4-a} g(x)dx$$

Let $f(x) = \int_0^a g(x)dx + \int_0^{4-a} g(x)dx$

$$f'(x) = g(a) - g(4-a)$$

Since $\frac{dg(x)}{dx} > 0$ $g(x)$ is $\uparrow$ function

$a + b = 4$ and $a < 2$, $a < 4 - a$

$$g(a) < g(4-a), \quad \frac{d(\phi(a))}{da} < 0$$

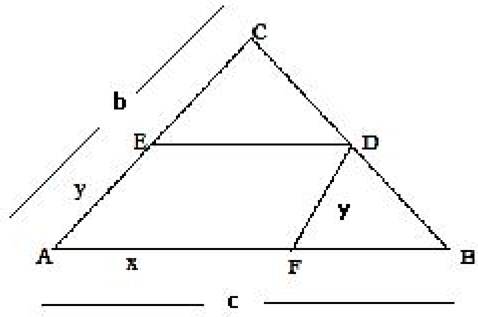
$$\frac{d(\phi(a))}{d(b-a)} \times \frac{d(b-a)}{da} < 0$$

$$\frac{d(\phi(a))}{d(b-a)} \times -2 < 0$$

$$\frac{d(\phi(a))}{d(b-a)} > 0$$

$\therefore \phi(a) \uparrow$ as $b - a$ increases

26.



from similar Δs FBD and ABC,

$$\frac{c-x}{c} = \frac{y}{b} \ or \ y = \left(\frac{b}{c}\right)(c-x)$$

$\Rightarrow Z =$ Area of AFDE =

$$xy\sin A = \frac{b\sin A}{c}(cx - x^2) \qquad (1)$$

0<x<c

$$\frac{dZ}{dx} = \frac{b\sin A}{c}(c-2x) = 0 \Rightarrow x = \frac{c}{2}$$

$$\left(\frac{d^2Z}{dx^2}\right)_{x=c/2} = \frac{-2}{c}b\sin A < 0$$

$\Rightarrow$ Z has maxima as the only extrema, so the greatest area of parallelogram AFDE.

$$= \left(\frac{b}{c}\right)\sin A\frac{c^2}{4} = \frac{1}{2}\Delta ABC$$

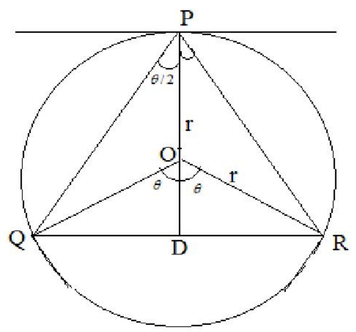
$$= \frac{1}{2} \times \frac{1}{2}\begin{vmatrix} p^2 & -p & 1 \\ q^2 & q & 1 \\ r^2 & r & 1 \end{vmatrix}$$

$R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$

$$=\frac{1}{4}\begin{vmatrix} p^2 & -p & 1 \\ q^2-p^2 & q+p & 0 \\ r^2-p^2 & -r+p & 0 \end{vmatrix}$$

$$=\frac{1}{4}(q+p)(q+r)(p-r)$$

27.



Let O be the centre r the radius of the circle. Let QR be the chord parallel to the tangent at the point p on the circle.

Let $\angle QPR=\theta,\ \therefore \angle QOD=\angle ROD=\theta;$

Area of ΔPQR

$$=A=\frac{1}{2}QRPD=QD(OP+OD)$$

$$=r\sin\theta(r+r\cos\theta)$$

$$=\frac{1}{2}r^2(2\sin\theta+\sin 2\theta),\ 0<\theta\le\pi/2$$

$$\Rightarrow\frac{dA}{d\theta}=r^2(\cos\theta+\cos 2\theta),$$

and $\Rightarrow\dfrac{d^2A}{d\theta^2}=-r^2(\sin\theta+2\sin 2\theta)$

For max. or min. of A, $\dfrac{dA}{d\theta}=0$

$$\Rightarrow\cos 2\theta+\cos\theta=0$$

$$\Rightarrow 2\cos^2\theta+\cos\theta-1=0$$

$$\Rightarrow(2\cos\theta-1)(\cos\theta+1)=0$$

$$\Rightarrow\cos\theta=\frac{1}{2}\quad or\quad \cos\theta=-1$$

$$\Rightarrow\cos\theta=\frac{\pi}{3}(\because\ 0<\theta<\pi/2)$$

when $\theta=\dfrac{\pi}{3},\ \dfrac{d^2A}{d\theta^2}=-\dfrac{3\sqrt{3}}{2}(-ve)$

$\Rightarrow$ A is max. when $\theta=\dfrac{\pi}{3}$, the only critical point. Thus, max. (greatest) area

$$\Rightarrow A=\frac{1}{2}r^2\left[2\sin\left(\frac{\pi}{3}\right)+\sin\left(\frac{2\pi}{3}\right)\right]=\frac{1}{4}(3\sqrt{3})r^2$$

28. Applying $R_3\to R_3-R_1-2R_2$

we get $f'(x)=\begin{vmatrix} 2ax & 2ax-a & 2ax+b+1 \\ b & b+1 & -1 \\ 0 & 0 & 1 \end{vmatrix}$

$$=\begin{vmatrix} 2ax & 2ax-1 \\ b & b+1 \end{vmatrix}=\begin{vmatrix} 2ax & -1 \\ b & 1 \end{vmatrix}\qquad [C_2\to C_2-C_1]$$

$$\Rightarrow f'(x)=2ax+b$$

Integrating, we get $f(x)=ax^2+bx+c$

where c is an arbitary constant.

since f has a maximum at x=5/2

$f'(5/2)=0\Rightarrow 5a+b=0$ (1)

Also $f(0)=2\Rightarrow c=2$

and $f(1)=1\Rightarrow a+b+c=1$

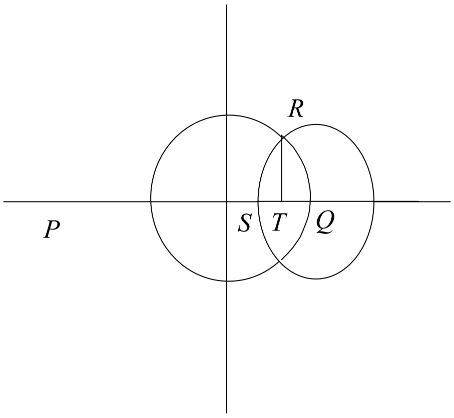
$\therefore a+b=-1$ (2)

solving (1) and (2) for a,b we get,

$$a=\frac{1}{4},\ b=\frac{-5}{4}$$

thus, $f(x)=\dfrac{1}{4}x^2-\dfrac{5}{4}x+2$.

29.



$x^2+y^2=1......(1)$, $Q(1,0)$

Let radius O is variable circle be r.

$(x-1)^2+y^2=r^2.......(2)$

Subsitute (2) from (1)

$$2x-1=1-r^2,\ x=\frac{1-r^2}{2}=OT$$

$$RT=\sqrt{OR^2-OT^2}=\sqrt{1-\left[1-\frac{r}{2}\right]^2}$$

area of ΔQSR is $A=\frac{1}{2}QS.RT$,

$$A^2=\frac{1}{4}(QS)^2\left(RT^2\right)$$

$$A^2=\frac{1}{4}r^2\left(r^2-\frac{r^2}{4}\right)$$

$$A^2=\frac{1}{16}r^2\left(4r^4-r^6\right)$$

$$\frac{d(A)^2}{dr}=\frac{1}{16}\left(16r^3-6r^5\right)=0,\ r=2\sqrt{\frac{2}{3}}$$

$$\frac{d^2\left(A^2\right)}{dt^2}=\frac{1}{16}\left(48r^2-30r^4\right)=-\frac{16}{3}<0$$

$$\therefore\ A_{\max}=\frac{4}{3\sqrt{3}}$$

30.

$$\frac{dD}{dx}=\frac{1}{300}\left(\frac{900}{x}+x\right)500\times0.4+1.5\times\frac{500}{x}$$

$$\frac{5}{3}\left(\frac{-900}{x^2}+1\right)\times0.4-\frac{750}{x^2}=0$$

$$\Rightarrow\frac{2}{3}\left(-\frac{900}{x^2}+1\right)=\frac{750}{x^2}$$

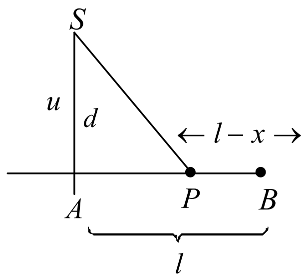
$$\Rightarrow\frac{2}{3}\left(x^2-900\right)=750$$

$$\Rightarrow 2x^2-1800=2250$$

$$\Rightarrow 2x^2=4050\Rightarrow x^2=2025$$

$$\Rightarrow x=45\ kmph$$

31.



$$sp=\sqrt{d^2+x^2}$$

$$t_1=\sqrt{a^2+x^2}\qquad t_2=\frac{L-x}{v}$$

$$T=\sqrt{\frac{d^2+x^2}{u}}+\frac{L-x}{v}$$

$$\frac{dT}{dx}=\frac{x}{u\sqrt{d^2+x^2}}-\frac{1}{v}$$

$$\frac{dT}{dx}=0,\ v^2x^2=u^2d^2+u^2x^2$$

$$x=\frac{ud}{\sqrt{v^2-u^2}}$$

32. $y=x-x^3$ $\qquad y'=1-3x^2$

equation of tangent $p(x_1,y_1)$

$$y-y_1=\left(1-3x_1^2\right)(x-x_1)$$

meet curve again at (x_2,y_2)

$$x_2-x_3^3-\left(x_1-x_1^3\right)=\left(1-3x_1^2\right)(x_2-x_1)$$

$$(x_2-x_1)\left[1-\left(x_2^2+x_1x_2+x_1^2\right)\right]=$$

$$(x_2-x_1)\left(1-3x_1^2\right)$$

$$x_2^2+x_1x_2-2x_1^2=0$$

$$\left[x_2+\frac{x_1}{2}\right]^2=\frac{9x_1^2}{4},\quad x_2+\frac{x_1}{2}=\pm\frac{3x_1}{2}$$

$x_1=x_2$ (not possible) $\therefore\ x_2=-2x_1$

Q is $\left(-2x_1,-2x_1+8x_1^3\right)$

$L_1(\alpha,\beta)$ point of trisection of PQ

$\alpha=\dfrac{|2x_1-2x_1|}{3}=0$ hence L_1 lies on y-axis

$L_2(h,k)$ is other point

$$h=\frac{x_1-4x_1}{3}=-x_1\ ,\ k=\frac{y_1-4x_1+16x_1^3}{3}$$

$$k=\frac{x_1-x_1^3-4x_1+16x_1^3}{3}=-x_1+5x_1^3$$

$$y=x-5x^3$$

33. Let $p\equiv(r\cos\theta,r\sin\theta)$

$$5r^2\cos^2\theta-8r^2\sin^2\theta\cos\theta+5r^2\sin^2\theta=4$$

$$5r^2-4r^2\sin 2\theta=4$$

$$r^2=\frac{4}{5-4\sin\theta}$$

$$r_{max}^2=4 \qquad r_{min}^2=\frac{4}{9}$$

$$r_{max}=2 \qquad r_{min}=\frac{2}{3}$$

clearly r is max for $\sin 2\theta=1\ \ \theta=\pi/4$

min for $\theta=-3\pi/4$

$\therefore$

$$\left[\pm2\cos\frac{\pi}{4},\pm2\sin\frac{\pi}{4}\right]\text{and}\left[\pm\frac{2}{3}\cos\frac{3\pi}{4},\pm\frac{2}{3}\sin\frac{3\pi}{4}\right]$$

$$\left[\pm\sqrt{2},\pm\sqrt{2}\right] \qquad \left[\pm\frac{\sqrt{2}}{3},\pm\frac{\sqrt{2}}{3}\right]$$

max $\qquad$ min

34.

$$f(x)=a_0x^6+a_1x^5+a_2x^4+a_3x^3+a_4x^2$$

$$+a_5x+a_6$$

$$\frac{f(x)}{x^3}=a_0x^3+a_1x^2+a_2x+a_3$$

$$+\frac{a_4}{x}+\frac{a_5}{x^2}+\frac{a_6}{x^3}$$

$$\lim_{x\to0}\left\{1+\frac{f(x)}{x^3}\right\}^{1/x}=e^2$$

$$e^{\lim\limits_{x\to0}\frac{1}{x}\log\left(1+\frac{f(x)}{x^3}\right)}=e^2$$

$$\lim_{x\to0}\frac{1}{x}\log\left(1+\frac{f(x)}{x^3}\right)=2$$

$$\lim_{x\to0}\log\frac{1+a_0x^3+a_1x^2+a_2x+a_3+\frac{a_4}{x}+\frac{a_5}{x^2}+\frac{a_6}{x^3}}{x}=2$$

Lt is $\propto$ at $a_3 = a_4 = a_5 = a_0 = 0$

$f(x) = a_0x^6 + a_1x^5 + a_2x^4$

$$\lim_{x\to 0} \frac{\log\left(1+\left(a_0x^3 + a_1x^2 + a_2x\right)\right)}{x} = 2$$

$f(x) = a_0x^6 + a_1x^5 + 2^{44}$

$f'(x) = 6a_0x^5 + 5a_1x^4 + 8x^3$

$f'(0) = 0$, $f'(2) = 0$, $f'(\phi) = 0$

$f'(0) = 0 \Rightarrow a_0, a_1$ have any real value

$f'(2) = 0 \Rightarrow 6\times 32a_0 + 5\times 6a_0 + 8\times 8 = 0$

$12a_0 + 5a_0 + 4 = 0$

$f'(1) = 0 \quad 6a_0 + 5a_1 + 8 = 0$

$\Rightarrow 6a_0 - 4 = 0 \quad a_0 = {}^2\!/_3, a_1 = -{}^{12}\!/_5$

$\Rightarrow f(x) = \frac{2}{3}x^6 - \frac{12}{5}x^5 + 2x^4$

35. Let Length of the side of base be x cm and y be the perpendicular height of the pyramid slant height be l.

Volume $= \frac{1}{3}\times$ (base area) $\times$ height

$v = \frac{1}{3}x^2y = 36\sqrt{2}$ (given), $\quad y = \frac{108\sqrt{2}}{x^2}$

Surface area $= \frac{1}{2}\times$ (perimeter of base) $\times$

slant height $= \left(\frac{1}{2}\right)(4n)l$

From ERG, $l = \sqrt{\frac{x^2}{4} + y^2}$

$s = 2x\sqrt{\frac{x^2}{4} + y^2} = \sqrt{x^4 + 4x^2y^2}$

$s = \sqrt{x^4 + 4x^2\left(\frac{108\sqrt{2}}{x^2}\right)^2}$

$s(x) = \sqrt{x^4 + \frac{8(108)^2}{x^2}}$

Let $f(x) = x^4 + \frac{8(108)^2}{x^2}$

$f'(x) = 4x^3 - \frac{16(108)^2}{x^3} = 0$

at $x = 6$ which is point of minima

Hence $x = 6cm \quad y = 3\sqrt{2}$

$l = \sqrt{9+36} = \sqrt{45} \Rightarrow l = 3\sqrt{5}$

36. Any root t_0 to this equation gives two real & distinct value of x if $|t_0| > 2$.

Thus, we need to find conditions in t we have 3 real and distinct roots.

$f(t) = 2t^3 - 9t^2 + 30 - a$

$f'(t) = 6t^2 - 18t = 0 \Rightarrow t = 0,3$

$f(t) = 0$ has 3 real & distinct root if

$f(0).f(3) < 0$

$(30-a)(54-81+30-a) < 0$

$(30-a)(3-a) < 0$

$a \in (3,30)$

none of roots should lie in $[-2,2]$

$f(-2) > 0 \qquad f(2) > 0$

$(-16-36+30-a) > 0 \quad (16-36+30-a) > 0$

$a + 22 < 0 \qquad a - 10 < 0$

$a < -22 \qquad a < 10$

$a < -22$

but $a \in (3,30)$

$\therefore$ no solution

37. Since minimum occurs before maximum

$a < 0$

Since a is a root of $x^2 - x - 6 = 0$

$x = -2$

$g(x) = ax^3 + bx^2 + cx + d$

$g(x) = -2x^3 + bx^2 + cx + d$

$g(x) = -6x^2 + 2bx + c$

$= -6(x+2)(x-2)$

$b = 0 \qquad c = 24$

since maximum & minimum values are non zero $g(-2)$ & $g(2)$ are $+ve$

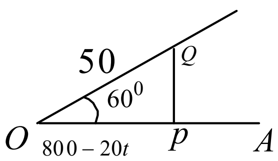
$g(x) = -2x^3 + 24x + d$

$g(-2) > 0 \Rightarrow 16 - 48 + d > 0 \qquad d > 32$

$g(2) > 0 \Rightarrow -16 + 48 + d > 0 \qquad d > -32$

$a = -2,\ b = 0,\ c = 24,\ d > 32$

38.



After t sec

So $AP = 20t \qquad OQ = 5t$

$OP = 800 - 20t$

In ΔOPQ

$PQ^2 = OP^2 + OQ^2 - 20POQ\cos 60$

$= (800-20t)^2 + 25t^2 - 2(800-20t)5t(-1/2)$

$= (800-20t)^2 + 25t^2 - 5(800t - 20t)$

$f(t) = PQ^2 = (800-20t)^2 + 25t^2 - 5(800t - 20t^2)$

$f(t) = 2(800-20t)(-20) + 50t$

$-5(800-40t) = 1050t - 36000$

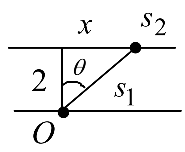
$f'(t) = 1050\left[t - \dfrac{240}{7}\right]$

$f'(t) = 0 \quad f'(t) = 1050\left[t - \dfrac{240}{7}\right] = 0$

$t = \dfrac{240}{7} \qquad f''(t) = tx$

f has min at $t = \dfrac{240}{7}$ sec

39.



$T = t_1 + t_2 = \dfrac{s_1}{2} + \dfrac{s_2}{5}$

$s_1 = \dfrac{2}{\cos\theta},\ s_2 = 6 - 2\tan\theta$

$\Rightarrow T = \dfrac{1}{\cos\theta} + \dfrac{6 - 2\tan\theta}{5}$

$T' = \sec\theta\tan\theta - \dfrac{2}{5}\sec^2\theta = 0$

$= \sec\theta(\tan\theta = 2/5\sec\theta)$

$\dfrac{\sin\theta}{\cos\theta} = \dfrac{2}{5\cos\theta}$

$\sin\theta = 2/5, \cos\theta = \dfrac{\sqrt{21}}{5}$

$\Rightarrow 2\tan\theta = 2 \times \dfrac{2}{\sqrt{21}} = 4/\sqrt{21}$

$\therefore\ k = 4/\sqrt{21}$ miles

40. Case - I :- $x < -2$

$f(x) = |-(x+2) + 2(x-2)| - x$

$=|-x-2+2x-4|-x=|x-6|-x$

$=-(x-6)-x=-2x+6$

Case - II :- $-2\le x<0$

$f(x)=|(x+2)+2(x-2)|-x$

$=|3x-2|-x=-4x+2$

Case - III :- $0\le x<2$

$f(x)=|x+2+2(x-2)|+x$

$=|3x-2|+x$

$$=\begin{cases}-(3x-2)+x, & a\le x<2/3\\ (3x-2)+x, & \dfrac{2}{3}\le x<2\end{cases}$$

$$f(x)=\begin{cases}-2x+2, & 0\le x<2/3\\ 4x-2, & \dfrac{2}{3}\le x<2\end{cases}$$

Case - IV :- $f(x)=|x+2-2(x-2)|+x$

$=|-x+6|+x$

$$=\begin{cases}-(x-6)+x, & 2\le x<6\\ (x-6)+x, & x\ge 6\end{cases}$$

$$=\begin{cases}6 & 2\le x<6\\ 2x-6 & x\ge 6\end{cases}$$